PHYSICS 1B · MIDTERM 1 CHEATSHEET

① Measuring density | ② Flow rate & Bernoulli | ③ U(x) & equilibrium | ④ Damped (+driven) oscillations | ⑤ Standing waves | ★ Other ch.1–3 | ∑ Math

5 Qs · 100 min = 20 min each. Hard one? Skip and come back. Write the law/ODE first, **label every middle step** (rubric), explain your logic in words, box the answer, then **check limits** ($b\to 0$, $\rho_{air}\to 0$, $A_1=A_2$) and **units**. No calculator, so answers can stay symbolic.

① MEASURING DENSITY · BUOYANCY

review p.2 · ch.1

Fluid statics basics

$\rho = \frac{m}{V}$, $P = \frac{F}{A}$ (Pa = N/m²). At rest: $\boxed{P = P_0 + \rho g h}$.

Depends **only on depth**, not on container shape. Gauge = $P - P_{atm}$.

Pascal: a pressure change spreads to the whole fluid.

Hydraulic lift: $\frac{F_1}{A_1} = \frac{F_2}{A_2}$ but $d_2 = d_1\frac{A_1}{A_2}$, so work in = work out.

Archimedes: $\boxed{F_B = \rho_{fl} V_{disp}\, g}$ = weight of displaced fluid.

Where it comes from: $F_B = (P_{bot} - P_{top})A = \rho_{fl} g h A$. It does **not** depend on depth or on what the object is made of.

Weighing method (lecture: air vs. submerged)

(a) in air: $T = w = \rho_{obj} V_{obj}\, g$ **(b) submerged** ($V_{disp} = V_{obj}$): $T = w_{app} = w - F_B$

$\Rightarrow F_B = w - w_{app} = \rho_{fl} V g$. Divide (a)/(this):

$$\boxed{\frac{\rho_{obj}}{\rho_{fl}} = \frac{w}{w - w_{app}}} \quad \text{also } V = \frac{w - w_{app}}{\rho_{fl} g}$$

Unknown liquid: weigh one object in water and in the liquid: $\frac{\rho_{liq}}{\rho_w} = \frac{w - w_{app,liq}}{w - w_{app,w}}$.

★ His slide question: "What if we DON'T ignore air density?" Air also buoys the object:

$w_{air} = (\rho_{obj} - \rho_{air})Vg$, $w_{liq} = (\rho_{obj} - \rho_{fl})Vg$

Subtract: $w_{air} - w_{liq} = (\rho_{fl} - \rho_{air})Vg$. Divide:

$$\frac{\rho_{obj} - \rho_{air}}{\rho_{fl} - \rho_{air}} = \frac{w_{air}}{w_{air} - w_{liq}}$$

$$\boxed{\rho_{obj} = \rho_{air} + (\rho_{fl} - \rho_{air})\frac{w_{air}}{w_{air} - w_{liq}}}$$

Check: $\rho_{air}\to 0$ gives the lecture result. Error $= \rho_{air}(1 - \frac{w}{w - w_{app}}) < 0$, so ignoring air slightly **overestimates** ρ_{obj}. It's tiny, since $\rho_{air} \approx 1.2$ vs $\rho_w = 1000$ kg/m³ (~0.1%).

Floating & FBD tricks

- Floats: $F_B = w \Rightarrow \frac{V_{sub}}{V} = \frac{\rho_{obj}}{\rho_{fl}}$ (ice 0.92, so ~90% of an iceberg is under). Sinks iff $\rho_{avg} > \rho_{fl}$. Hollow: $\rho_{avg} = m_{tot}/V_{tot}$. Two layers: $\rho_{obj}V = \rho_1 V_1 + \rho_2 V_2$.
- Object hung in a beaker on a scale: $T + F_B = mg$. The scale reads $W_{beaker+water} + F_B$ (Newton's 3rd law: object pushes the water down).
- Melting ice: level unchanged. Rock tossed from boat: level **falls**.

② FLOW RATE · CONTINUITY · BERNOULLI

review p.3

Volume flow rate $\boxed{Q = \frac{dV}{dt} = \frac{d}{dt}(Ax) = A\frac{dx}{dt} = Av}$ [m³/s].

Mass flow $\dot m = \rho A v$.

Continuity (incompressible): $\boxed{Q_1 = Q_2 \Rightarrow A_1 v_1 = A_2 v_2}$

for any points 1, 2. $v \propto 1/A \propto 1/d^2$: halve d gives 4× v.

Branches: $Q_{in} = \sum Q_{out}$.

Bernoulli (incompressible, frictionless/non-viscous, steady, along a streamline):

$$\boxed{P + \tfrac{1}{2}\rho v^2 + \rho g y = \text{const}}$$

Each term is an energy per volume (J/m³ = Pa): pressure + KE density + PE density.

Derivation (work–energy process): push a slug dV through: $W = P_1 A_1 dx_1 - P_2 A_2 dx_2 = (P_1 - P_2)dV$ (continuity: $A_1 dx_1 = A_2 dx_2 = dV$). $\Delta K = \frac{1}{2}\rho dV(v_2^2 - v_1^2)$, $\Delta U = \rho dV g(y_2 - y_1)$. Then $W = \Delta K + \Delta U \Rightarrow P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) + \rho g(y_2 - y_1)$.

Limits: v=0 gives statics $P = P_0 + \rho g h$. Same height: **faster means lower P** (Venturi).

Applications

- **Torricelli** (open tank, hole depth h, $A_{tank} \gg A_{hole}$): $P_1 = P_2 = P_{atm}$, $v_1 \approx 0$ give $\boxed{v = \sqrt{2gh}}$ (free fall). $Q = A_h\sqrt{2gh}$. Hole at height H: range $2\sqrt{hH}$.
- **Venturi** (horizontal, $A_1 > A_2$): $P_1 - P_2 = \frac{1}{2}\rho v_1^2[(\frac{A_1}{A_2})^2 - 1]$. Manometer: $P_1 - P_2 = (\rho_m - \rho)gh$.
- **Prandtl/Pitot tube** (disc. wk 2): front opening 1 is a dead spot, $v_1 = 0$. Side opening 2 sees air at speed v_2. Bernoulli: $P_1 - P_2 = \frac{1}{2}\rho_{air} v_2^2$. U-tube: $\rho g h = P_1 - P_2$ gives $\boxed{h = \frac{\rho_{air} v_2^2}{2\rho g}}$ so $v_2 = \sqrt{\frac{2\rho g h}{\rho_{air}}}$.
- **Lift:** faster air on top means lower P on top: $F \approx \frac{1}{2}\rho(v_t^2 - v_b^2)A$ (roofs in storms, curveball).
- **Siphon:** exit speed $\sqrt{2gd}$ (d = drop from surface to exit). Pipe rising at constant A: v is constant, so P drops by $\rho g \Delta y$.

Viscous (Bernoulli fails): $Q = \frac{\pi R^4 \Delta P}{8\eta L}$ (R/2 gives Q/16). Turbulent if $Re = \frac{\rho v D}{\eta} \gtrsim 2000$.

★ SHM & PENDULUMS

ch.2 · disc. wk 2

$F = -kx \Rightarrow \ddot x = -\frac{k}{m}x \Rightarrow \boxed{x = A\cos(\omega t + \phi)}$, $\omega = \sqrt{k/m} = 2\pi f = \frac{2\pi}{T}$.

$v = -A\omega\sin(\omega t + \phi)$, $a = -\omega^2 x$. $v_{max} = A\omega$ (at x=0), $a_{max} = A\omega^2$ (at ±A), $v(x) = \pm\omega\sqrt{A^2 - x^2}$.

ICs: $A = \sqrt{x_0^2 + (v_0/\omega)^2}$, $\tan\phi = -\frac{v_0}{\omega x_0}$. **Energy** $E = \frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}mv_{max}^2$. KE = PE at $x = A/\sqrt{2}$.

Vertical spring: same ω, equilibrium shifted by mg/k. Springs in parallel: $k_1 + k_2$; in series: $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$.

Simple pendulum: $\tau = -mgL\sin\theta = I\alpha$, $I = mL^2$, so $\ddot\theta = -\frac{g}{L}\sin\theta \approx -\frac{g}{L}\theta$ (small angle).

$\boxed{\omega = \sqrt{g/L},\ T = 2\pi\sqrt{L/g}}$, independent of m and A.

Measure g: $g = 4\pi^2 L/T^2$. For better precision: long L, time many periods, small amplitude. Clock: $\frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta L}{L}$, so 2% too long means 1% slow = 14.4 min/day.

Physical pendulum (d = pivot→CM): $I\ddot\theta = -mgd\sin\theta$, so

$\boxed{\omega = \sqrt{\frac{mgd}{I}},\ T = 2\pi\sqrt{\frac{I}{mgd}}}$ Rod pivoted at end: $T = 2\pi\sqrt{\frac{2L}{3g}}$.

I: rod (center) $\frac{ML^2}{12}$, rod (end) $\frac{ML^2}{3}$, hoop MR^2, disk $\frac{1}{2}MR^2$, sphere $\frac{2}{5}MR^2$, shell $\frac{2}{3}MR^2$. Parallel axis: $I = I_{cm} + Md^2$.

Torsional: $\tau = -\kappa\theta$, $\omega = \sqrt{\kappa/I}$, $T = 2\pi\sqrt{I/\kappa}$ (κ in N-m/rad).

④ DAMPED OSCILLATIONS: THE FULL PROCESS

review p.5 · lect. 4 · "follow the process"

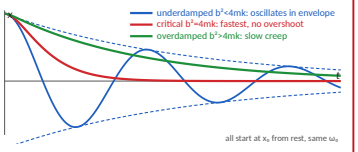
Setup: drag $F = -bv$ (b in kg/s). Newton: $ma = -bv - kx$, so

$$\boxed{m\ddot x + b\dot x + kx = 0}\ (1) \iff \ddot x + 2\gamma\dot x + \omega_0^2 x = 0$$

$\gamma = \frac{b}{2m}$ (decay rate, 1/s), $\omega_0 = \sqrt{\frac{k}{m}}$ (natural ang. freq.).

Damping ratio $\zeta = \frac{\gamma}{\omega_0} = \frac{b}{2\sqrt{mk}}$.

Which case? Compare $\boxed{b^2 \text{ vs } 4mk} \iff \gamma \text{ vs } \omega_0 \iff b \text{ vs } 2\sqrt{mk}$.



	UNDERDAMPED $b^2 < 4mk$	CRITICAL $b^2 = 4mk$	OVERDAMPED $b^2 > 4mk$
roots	complex: $\gamma \pm i\omega$	repeated: $r = \gamma$	2 real: $r_{1,2} = \gamma \pm \alpha$
x(t)	$A_0 e^{-bt/2m}\cos(\omega t + \phi)$	$(A + Bt)\,e^{-\gamma t}$	$C_1 e^{-r_1 t} + C_2 e^{-r_2 t}$
freq.	$\omega = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} = \sqrt{\omega_0^2 - \gamma^2} < \omega_0$	$\omega = 0$: no oscillation!	$\alpha = \frac{\sqrt{b^2 - 4mk}}{2m} = \sqrt{\gamma^2 - \omega_0^2}$
motion	oscillates in a decaying envelope $\pm A_0 e^{-\gamma t}$; ∞ zero crossings; $T_d = 2\pi/\omega > T_0$	**quickest** return to equilibrium with no overshoot (≤1 crossing)	slow creep back ("too much resistance"), ≤1 crossing; late-time ~ $e^{-(\gamma-\alpha)t}$
e.g.	guitar string, swing, small b	car shocks, door closer, meter needle	mass in honey, very stiff damper

A · UNDERDAMPED derivation (lecture)

1. **Try** $x = A_0 e^{-rt}\cos(\omega t + \phi)$. Let $\theta = \omega t + \phi$.
2. $\dot x = -A_0 e^{-rt}(r\cos\theta + \omega\sin\theta)$
3. $\ddot x = A_0 e^{-rt}[(r^2 - \omega^2)\cos\theta + 2r\omega\sin\theta]$
4. **Plug into (1):** $A_0 e^{-rt}\{[m(r^2 - \omega^2) - br + k]\cos\theta + (2mr - b)\omega\sin\theta\} = 0$
5. True for **all** t, and cos θ, sin θ are independent, so **each coefficient = 0:**

   sin θ: $2mr - b = 0 \Rightarrow \boxed{r = \frac{b}{2m}}$

   cos θ: $m(r^2 - \omega^2) - br + k = 0$ (2)
6. Put r into (2): $\boxed{\omega^2 = \frac{k}{m} - \frac{b^2}{4m^2}}$. ω is real only if $b^2 < 4mk$.
7. $x = \underbrace{A_0 e^{-bt/2m}}_{\text{decaying amp.}}\underbrace{\cos(\omega t + \phi)}_{\text{oscillation}}$.

Check b→0: SHM ✓.

B · OVERDAMPED derivation (lecture)

1. **Try** $x = A_0 e^{-rt}$: $\dot x = -rx$, $\ddot x = r^2 x$.
2. **Plug in:** $(mr^2 - br + k)A_0 e^{-rt} = 0$ (slide typo "kr") so $mr^2 - br + k = 0$
3. Quadratic formula: $\boxed{r = \frac{b \pm \sqrt{b^2 - 4mk}}{2m} = \gamma \pm \alpha}$
4. $b^2 > 4mk$ gives two real roots $r_1 = \gamma + \alpha$ (fast), $r_2 = \gamma - \alpha$ (slow, dominates at late t). Both > 0 since $\alpha < \gamma$, so x always decays.
5. Linear ODE, so superpose: $\boxed{x = C_1 e^{-r_1 t} + C_2 e^{-r_2 t}}$. r_1, r_2 are the **two decay time scales** $1/r_1, 1/r_2$.
6. Same thing: $x = e^{-\gamma t}[C_1 e^{-\alpha t} + C_2 e^{\alpha t}] = e^{-\gamma t}[A\cosh\alpha t + B\frac{\sinh\alpha t}{\alpha}]$, with $A = C_1 + C_2$, $B = \alpha(C_2 - C_1)$.

C · CRITICAL (b^2=4mk): "Not exactly!"

1. $r_1 = r_2 = r$, so $(C_1 + C_2)e^{-rt}$ has **only one constant**. It can't match both x_0 and v_0, so we need a 2nd solution.
2. **From over:** α→0 gives $\cosh\alpha t \to 1$, $\frac{\sinh\alpha t}{\alpha} \to t$
3. **From under:** $x = e^{-\gamma t}[A\cos\omega t + B\frac{\sin\omega t}{\omega}]$, ω→0 gives $\cos\omega t \to 1$, $\frac{\sin\omega t}{\omega} \to t$ (slide typo "sinh/α")
4. Both give $\boxed{x = (A + Bt)e^{-\gamma t}}$, $\gamma = \frac{b}{2m} = \sqrt{\frac{k}{m}}$
5. **Verify** $te^{-\gamma t}$: $\dot x = (1 - \gamma t)e^{-\gamma t}$, $\ddot x = (\gamma^2 t - 2\gamma)e^{-\gamma t}$, so $\ddot x + 2\gamma\dot x + \gamma^2 x = e^{-\gamma t}[\gamma^2 t - 2\gamma + 2\gamma - 2\gamma^2 t + \gamma^2 t] = 0$ ✓

Initial conditions: one master pattern. $x(0) = x_0$, $\dot x(0) = v_0$. Let $\boxed{B \equiv v_0 + \gamma x_0}$.

Under: $x = e^{-\gamma t}[x_0\cos\omega t + \frac{B}{\omega}\sin\omega t]$

Critical: $x = e^{-\gamma t}[x_0 + Bt]$

Over: $x = e^{-\gamma t}[x_0\cosh\alpha t + \frac{B}{\alpha}\sinh\alpha t]$

Why (lecture): $x = e^{-\gamma t}[C\cos\omega t + D\sin\omega t]$. At t=0, $C = x_0$. $v(0) = -\gamma C + \omega D \Rightarrow D = \frac{v_0 + \gamma x_0}{\omega}$, so D = (const)/ω and the ω→0 limit is finite. From rest ($v_0 = 0$): $B = \gamma x_0$, e.g. critical $x = x_0(1 + \gamma t)e^{-\gamma t}$.

Amp–phase form (under): $A_0 = \sqrt{x_0^2 + (B/\omega)^2}$, $\tan\phi = -\frac{B}{\omega x_0}$.

Over, direct: $C_1 + C_2 = x_0$, $-r_1 C_1 - r_2 C_2 = v_0$, so $C_1 = -\frac{v_0 + r_2 x_0}{r_1 - r_2}$, $C_2 = \frac{v_0 + r_1 x_0}{r_1 - r_2}$.

Trend as b grows: ω falls and decay speeds up until critical.

Past critical, return gets **slower:** $r_2 = \gamma - \alpha \to \frac{\omega_0^2}{2\gamma} = \frac{k}{b} \to 0$. So critical is the fastest.

Any oscillator: swap in ω_0^2 (g/L, mgd/I, κ/I) and 2γ (= b/m).

④ cont. · ENERGY · Q · DRIVEN

disc. wk 3 · lect. 4 p.5

Energy loss

$E = \frac{1}{2}m\dot x^2 + \frac{1}{2}kx^2$. Then $\frac{dE}{dt} = \dot x(m\ddot x + kx) = \dot x(-b\dot x)$, so $\boxed{\frac{dE}{dt} = -b\dot x^2 \le 0}$

Weak damping (γ≪ω_0): $E \approx \frac{1}{2}kA_0^2 e^{-2\gamma t} = E_0 e^{-bt/m}$.

Energy decays twice as fast as amplitude: $\tau_E = \frac{m}{b}$, $\tau_A = \frac{2m}{b}$.

Disc. wk 3, E(t) for the CRITICAL case: $x = e^{-\alpha t}(A_0 + A_1 t)$, α = b/2m.

$\dot x = e^{-\alpha t}[-\alpha(A_0 + A_1 t) + A_1]$. Critical means $\alpha^2 = \frac{k}{m}$, so $E = \frac{1}{2}m(\dot x^2 + \alpha^2 x^2)$:

$E = \frac{1}{2}me^{-2\alpha t}[2\alpha^2(A_0 + A_1 t)^2 + A_1^2 - 2\alpha A_1(A_0 + A_1 t)]$

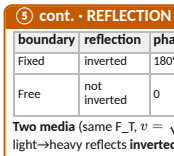
Quality factor $Q = \frac{\omega_0}{2\gamma} = \frac{m\omega_0}{b} = \frac{\sqrt{mk}}{b} \approx 2\pi\frac{E}{|\Delta E|_{cycle}}$. Critical: Q = ½. **Log decrement** $\ln\frac{x(t)}{x(t+T_d)} = \gamma T_d$. Amplitude halves at $t = \frac{2m\ln 2}{b}$.

Driven (forced) oscillator

$\boxed{m\ddot x + b\dot x + kx = F_0\cos\omega t}$, x = **transient** (the damped solution, dies out) + **steady state** $A\cos(\omega t - \delta)$. It ends up oscillating at the **driving** ω, not ω_0.

1. $\theta = \omega t - \delta$: $x = A\cos\theta$, $\dot x = -A\omega\sin\theta$, $\ddot x = -A\omega^2\cos\theta$
2. Plug in: $A(k - m\omega^2)\cos\theta - bA\omega\sin\theta = F_0\cos\omega t$
3. Expand: $\cos\theta = \cos\omega t\cos\delta + \sin\omega t\sin\delta$, $\sin\theta = \sin\omega t\cos\delta - \cos\omega t\sin\delta$
4. No sin ωt on the RHS: $(k - m\omega^2)\sin\delta - b\omega\cos\delta = 0$ gives $\boxed{\tan\delta = \frac{b\omega}{k - m\omega^2}}$ (phase lag)
5. cos ωt: $A[(k - m\omega^2)\cos\delta + b\omega\sin\delta] = F_0$. Use $\sin\delta = \frac{b\omega}{D}$, $\cos\delta = \frac{k - m\omega^2}{D}$:

$$\boxed{A = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (b\omega)^2}}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}$$



ω→0: $A \to F_0/k$, δ→0 (in phase).

ω=ω_0: $A = \frac{F_0}{b\omega_0}$, δ=90°.

ω≫ω_0: $A \to \frac{F_0}{m\omega^2}$, δ→180°.

Resonance at ω≈ω_0 (exact peak $\sqrt{\omega_0^2 - 2\gamma^2}$. Smaller b gives a taller, narrower peak (width ≈ 2γ = b/m).

⑤ cont. · REFLECTION · INTERFERENCE

lect. 6

boundary	reflection	phase	condition
Fixed	inverted	180°	$y(0,t) = 0 \Rightarrow y_r = -y_i$
Free	not inverted	0	$F_y = -F_T\frac{\partial y}{\partial x} = 0$: slope 0, so $g(u) = f(-u)$

Two media (same F_T, $v = \sqrt{F_T/\mu}$, μ↑ means v↓): light→heavy reflects **inverted** (like a fixed end); heavy→light reflects **upright**. Transmitted pulse is always upright. ★ $\frac{A_r}{A_i} = \frac{v_2 - v_1}{v_1 + v_2}$, $\frac{A_t}{A_i} = \frac{2v_2}{v_1 + v_2}$.

Superposition: $y_1 = A\sin(kx - \omega t)$, $y_2 = A\sin(kx - \omega t + \phi)$ gives $\boxed{y = 2A\cos\frac{\phi}{2}\sin(kx - \omega t + \frac{\phi}{2})}$, $A_R = 2A|\cos\frac{\phi}{2}|$. φ = 0, 2π: constructive (2A). φ = π: destructive (0). Path diff Δ gives φ = 2πΔ/λ. Waves pass through each other; particles don't.

⑤ STANDING WAVES

review p.6 · lect. 6

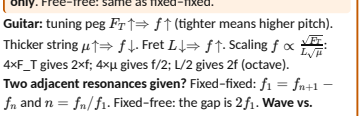
Two identical waves going opposite ways: $y_1 = A\sin(kx - \omega t)$, $y_2 = A\sin(kx + \omega t)$. Identity $\sin a + \sin b = 2\sin\frac{a+b}{2}\cos\frac{a-b}{2}$:

$$y(x,t) = \underbrace{2A\sin(kx)}_{\text{amp. depends on x}}\ \underbrace{\cos(\omega t)}_{\text{oscillation in time}}$$

Traveling $A\sin(kx - \omega t)$: space and time mixed, the pattern moves. Standing: **space and time separate, the pattern does not propagate.**

Nodes ($A_{local} = |2A\sin kx| = 0$): $kx = n\pi$, so $\boxed{x_n = n\frac{\lambda}{2}}$.

Antinodes (|sin kx|=1, amp. 2A): $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \ldots$ Node–node = λ/2, node–antinode = λ/4. Points in one loop move **in phase**; neighbouring loops are opposite. The string is flat everywhere when cos ωt = 0 (all energy kinetic then).



Fixed both ends (both ends nodes; only some λ fit):

$$\lambda_n = \frac{2L}{n},\quad f_n = \frac{n v}{2L} = \frac{n}{2L}\sqrt{\frac{F_T}{\mu}},\ \text{n = 1, 2, 3...}$$

n=1 is the fundamental (1st harmonic, λ = 2L). $f_n = nf_1$. Mode n has n loops, n antinodes, n+1 nodes (incl. ends).

Fixed–free (free end = antinode): $\lambda_n = \frac{4L}{n}$, $f_n = \frac{nv}{4L}$, **odd n only**. Free–free: same as fixed–fixed.

Guitar: tuning peg $F_T\uparrow \Rightarrow f\uparrow$ (tighter means higher pitch). Thicker string $\mu\uparrow \Rightarrow f\downarrow$. Fret $L\downarrow \Rightarrow f\uparrow$. Scaling $f \propto \frac{\sqrt{F_T}}{L\sqrt{\mu}}$: 4×F_T gives 2×f; 4×μ gives f/2; L/2 gives 2f (octave).

Two adjacent resonances given? Fixed–fixed: $f_1 = f_{n+1} - f_n$ and $n = f_n/f_1$. Fixed–free: the gap is $2f_1$. **Wave vs. string:** v is set by the string (F_T, μ); f is set by the source/mode; λ = v/f.

Energy in a standing wave

$\boxed{u = \frac{1}{2}\mu(\frac{\partial y}{\partial t})^2 + \frac{1}{2}F_T(\frac{\partial y}{\partial x})^2}$ (KE + stretch PE). For $y = A\sin kx\cos\omega t$, using $F_T k^2 = \mu\omega^2$ (since $v^2 = F_T/\mu = \omega^2/k^2$):

$$u = \tfrac{1}{2}\mu A^2\omega^2[\sin^2 kx\,\sin^2\omega t + \cos^2 kx\,\cos^2\omega t]$$

- **Node** (sin kx = 0): doesn't move, so KE density = 0. |cos kx| = 1, so the slope and PE density are **max**. Mostly PE.
- **Antinode**: max displacement and max speed (at different times); slope 0.
- L = integer × λ/2: $\int_0^L \sin^2 kx\,dx = \int_0^L \cos^2 kx\,dx = \frac{L}{2}$, so $\boxed{E = \frac{1}{4}\mu A^2\omega^2 L}$ **is constant in time** (KE and PE slosh locally).
- $E_n = \frac{1}{4}\mu L A_n^2\omega_n^2$, $\omega_n = \frac{n\pi v}{L}$, so $E_n \propto n^2$ at the same antinode amplitude.
- $P_{avg} = 0$: two equal and opposite traveling waves carry opposite energy fluxes. Energy is stored but not transported. Like a massive particle at rest ($E_0 = mc^2$, zero momentum); a traveling wave is like a massless one.

Why PE = ½F_T(∂y/∂x)²: $ds = \sqrt{dx^2 + dy^2} = dx[1 + \frac{1}{2}(\frac{\partial y}{\partial x})^2]$, extra length $d\ell = \frac{1}{2}(\frac{\partial y}{\partial x})^2 dx$, work $F_T d\ell$. PE depends on **local slope**, not height.

③ POTENTIAL CURVES · EQUILIBRIUM

review p.4

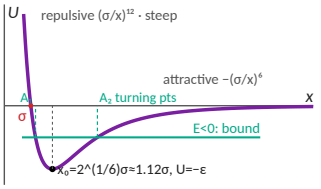
$\boxed{F(x) = -\frac{dU}{dx}}$: force points **downhill**. $U = -\int F\,dx$.

1st deriv.	force	2nd deriv. at x_0	type
$U' > 0$	$F < 0 \leftarrow$	$U'' > 0$ min	**stable:** F points back
$U' < 0$	$F > 0 \rightarrow$	$U'' < 0$ max	**unstable:** F points away
$U' = 0$	$F = 0$: equil.	$U'' = 0$ inflection	not stable (e.g. x^3)

Effective spring $k_{eff} = U''(x_0) = -\frac{dF}{dx}$ (cf. $F_s = -kx$). Taylor: $U \approx U(x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2$, so **small oscillations about any minimum are SHM:**

$\boxed{\omega = \sqrt{U''(x_0)/m}}$, angle coordinate: $\omega = \sqrt{U''(\theta_0)/I}$

Energy diagram: $K = E - U(x) \ge 0$, so the object is confined to $U \le E$. **Turning points** are where $U = E$ (v=0). Speed is max at the U minimum: $v = \sqrt{2(E - U)/m}$.



Lennard-Jones $\boxed{U = 4\epsilon[(\frac{\sigma}{x})^{12} - (\frac{\sigma}{x})^6]}$. The x^{-12} term is strongly repulsive (electron clouds overlap, Pauli exclusion). The $-x^{-6}$ term is attractive (fluctuating dipoles, van der Waals).

- $U(\sigma) = 0$. $\frac{dU}{dx} = 4\epsilon(-12\sigma^{12}x^{-13} + 6\sigma^6 x^{-7}) = 0$ gives $-2\sigma^6 + x_0^6 = 0$, so $\boxed{x_0 = 2^{1/6}\sigma \approx 1.12\sigma}$
- $U(x_0) = 4\epsilon(\frac{1}{4} - \frac{1}{2}) = -\epsilon$: well depth = binding (dissociation) energy
- $U'' = 4\epsilon(156\frac{\sigma^{12}}{x^{14}} - 42\frac{\sigma^6}{x^8})$, so $U''(x_0) = \frac{72\epsilon}{\sigma^2} = \frac{72\epsilon}{2^{1/3}\sigma^2} \approx 57\frac{\epsilon}{\sigma^2}$ and $\omega = \sqrt{\frac{72\epsilon}{\mu\sigma^2}}$ (μ = reduced mass, m/2 for identical atoms)
- $F = \frac{24\epsilon}{x}[2(\frac{\sigma}{x})^{12} - (\frac{\sigma}{x})^6]$: repulsive for $x < x_0$. E≥0 means unbound. Asymmetric well (anharmonic): average x grows with E, i.e. thermal expansion.

Ex. $U = ax^2 - bx^3$: $U' = 0$ at $x = 0$ ($U'' = 2a > 0$, stable) and at $x = \frac{2a}{3b}$ ($U'' = -2a$, unstable). Pendulum $U = mgL(1 - \cos\theta)$: $U''(0) = mgL$, so $\omega = \sqrt{mgL/mL^2} = \sqrt{g/L}$.

Reduced mass · disc. wk 2: two masses + one spring (rest length y_0). $y = x_2 - x_1$, $F_1 = k(y - y_0)$, $F_2 = -k(y - y_0)$.

$\frac{d^2}{dt^2}(y - y_0) = \ddot x_2 - \ddot x_1 = \frac{F_2}{m_2} - \frac{F_1}{m_1} = -k(\frac{1}{m_1} + \frac{1}{m_2})(y - y_0)$

$$\boxed{\omega = \sqrt{k(\tfrac{1}{m_1} + \tfrac{1}{m_2})} = \sqrt{k/\mu}},\ \mu = \frac{m_1 m_2}{m_1 + m_2}.$$ Check: $m_2 \to \infty$ gives $\sqrt{k/m_1}$ ✓

★ TRAVELING WAVES

ch.3 · lect. 5 · disc. wk 3

Mechanical waves need a medium and carry **energy and momentum, not mass**. Transverse: medium moves ⊥ to propagation. Longitudinal: ∥. P-wave: longitudinal, faster, through solids and liquids. S-wave: transverse, slower, solids only.

$\boxed{y = A\sin(kx - \omega t + \phi)}$ moves **+x**; $kx + \omega t$ moves −x (constant phase gives x = vt + c). $k = \frac{2\pi}{\lambda}$, $\omega = \frac{2\pi}{T}$,

$\boxed{v = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}}$. Any $f(x - vt)$ works. Pulse: Δx = vΔt.

Particle velocity $v_y = \frac{\partial y}{\partial t} = -A\omega\cos(kx - \omega t)$, max $A\omega$ (≠ wave speed!). $a_y = -\omega^2 y$. At a crest the particle has $v_y = 0$ while the crest itself moves at v.

Wave eq. $\boxed{\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}}$. It's linear, so superposition holds. (A wave plus a 2ω wave going the other way still solves it with the same v, but has no single direction.)

String speed: $F_{net,y} = F_T[(\frac{\partial y}{\partial x})_2 - (\frac{\partial y}{\partial x})_1] = \mu\Delta x\frac{\partial^2 y}{\partial t^2}$.

Divide by $F_T\Delta x$, let Δx→0: $\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{F_T}\frac{\partial^2 y}{\partial t^2}$, so $\boxed{v = \sqrt{F_T/\mu}}$.

Generally $v = \sqrt{\frac{\text{restoring}}{\text{inertia}}}$; sound $v = \sqrt{B/\rho}$.

Energy/power: $\boxed{E_\lambda = \frac{1}{2}\mu A^2\omega^2\lambda,\ P_{avg} = \frac{1}{2}\mu A^2\omega^2 v} = \frac{1}{2}\omega^2 A^2\sqrt{\mu F_T}$. Scalings: 2×A gives 4×P; 2×f gives 4×P; 4×F_T gives 2×v and 2×P.

Disc. wk 3: 4×μ (A, ω, F_T fixed): v and λ halve, but E_λ and P **double** (∝√μ). 4×F_T: v and λ double, so E_λ and P double.

Intensity $I = P/\text{area}$. Sphere: $I = \frac{P}{4\pi r^2}$, so $A \propto \frac{1}{r}$. 2D circle: $I = \frac{P}{2\pi r}$, so $A \propto \frac{1}{\sqrt r}$.

∑ MATH TOOLBOX · NUMBERS

no calculator

$\sin a \pm \sin b = 2\sin\frac{a \pm b}{2}\cos\frac{a \mp b}{2}$; $\cos a + \cos b = 2\cos\frac{a+b}{2}\cos\frac{a-b}{2}$; $\cos a - \cos b = -2\sin\frac{a+b}{2}\sin\frac{a-b}{2}$

$\sin(a \pm b) = \sin a\cos b \pm \cos a\sin b$; $\cos(a \pm b) = \cos a\cos b \mp \sin a\sin b$; $\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$; $\langle\sin^2\rangle = \langle\cos^2\rangle = \frac{1}{2}$

$a\cos x + b\sin x = R\cos(x - \delta)$, $R = \sqrt{a^2 + b^2}$, $\tan\delta = \frac{b}{a}$

$\cosh x = \frac{e^x + e^{-x}}{2}$, $\sinh x = \frac{e^x - e^{-x}}{2}$, $(\cosh)' = \sinh$, $(\sinh)' = \cosh$, $\cosh^2 - \sinh^2 = 1$, $e^{\pm x} = \cosh x \pm \sinh x$, $e^{i\theta} = \cos\theta + i\sin\theta$

Small x: $\sin x \approx \tan x \approx x$, $\cos x \approx 1 - \frac{x^2}{2}$, $e^x \approx 1 + x$, $(1 + x)^n \approx 1 + nx$, $\sinh x \approx x$, $\cosh x \approx 1$, $\frac{\sin x}{x} \to 1$

$ar^2 + br + c = 0 \Rightarrow r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. $\frac{d}{dt}[e^{-\gamma t}f] = e^{-\gamma t}(f' - \gamma f)$

#: √2 = 1.414, √3 = 1.732, √5 = 2.236, $\pi^2 \approx 9.87 \approx g$, $2^{1/6} = 1.122$, $2^{1/3} = 1.26$, e = 2.718, 1/e = 0.368, ln 2 = 0.693. g = 9.8 m/s². ρ: water 1000, sea 1025, ice 917, air ≈ 1.2, Hg 13 600 kg/m³. 1 atm = 1.013×10⁵ Pa = 760 mmHg.

Units: b kg/s · k N/m = kg/s² · κ N-m/rad · μ kg/m · Q m²/s · η Pa-s · γ, ω s⁻¹ · u J/m.